

CausalCache: Learning to Use and Select Visual History for Long-Horizon GUI Agents

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Abstract

Long-horizon GUI agents face two coupled bottlenecks under bounded visual context: a policy trained mainly for the current screen may not know how to use archived screenshots, while a memory selector cannot learn meaningful values from a policy that treats history as noise. We introduce CAUSALCACHE, a policy-preserving visual-memory interface for frozen GUI agents. Its history-gated key/value (HGKV) adapter adds low-rank residuals only to restored history-image tokens in the final eight language-model layers and is structurally bypassed when no history image is present, exactly preserving the base policy at zero visual-memory budget. We train this interface on GUI-Odyssey with correct, absent, shuffled, and irrelevant histories, without ordinary action behavior cloning. On top of the adapted policy, CAUSALCACHE defines success-anchored set utility as the change in likelihood of actions from successful trajectories when a subset of historical events is restored. A two-stage selector first learns singleton values from counterfactual HGKV readouts, then predicts coalition-conditioned marginal gains and a STOP decision to handle redundancy and complementarity under at-most- B budgets. All adaptation and model selection use GUI-Odyssey only; the same checkpoints are evaluated zero-shot on AndroidWorld and OS-World. On held-out GUI-Odyssey, HGKV improves correct-history over no-history likelihood by 0.1372 nats per token, reverses negative sensitivity to shuffled and irrelevant histories, and preserves no-history logits exactly. **[Pending: Insert final zero-shot AndroidWorld and OSWorld closed-loop improvements.]**

Introduction

GUI agents accumulate screenshots, actions, text arguments, coordinates, and interface changes over many decisions. Retaining every screenshot is expensive and can distract a multimodal policy; text-only trajectories discard toggle state, layout, transient values, and spatial correspondence. A practical agent must therefore decide which historical events remain available in high fidelity (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026).

Most memory controllers address only the selection problem. They retain recent events, retrieve visually similar

screenshots, or learn salience from task supervision (Zeng et al. 2026; Shi et al. 2026; Liu et al. 2026). These approaches implicitly assume that the action policy already understands multiple historical screenshots. Our experiments expose a more basic failure: a strong frozen GUI policy can change its output when history images are added while remaining insensitive, or even oppositely sensitive, to whether those images contain the correct historical evidence. In that regime, a better selector has little opportunity to improve executable behavior because the selected evidence is not reliably consumed.

Naively fine-tuning the whole policy is not a clean solution. Full-layer adaptation simultaneously changes current-screen understanding, action syntax, coordinate grounding, and termination behavior. It can therefore improve teacher-forced history sensitivity while overwriting the cross-platform action prior that made the base policy useful in the first place. The desired module should instead satisfy three behavioral conditions:

1. with no restored screenshot, it exactly recovers the frozen policy;
2. with irrelevant or corrupted history, it learns to ignore the added evidence;
3. with correct and useful history, it may change the action distribution toward successful behavior.

CAUSALCACHE implements this separation in two stages. First, a history-gated KV adapter modifies only the key and value projections of historical image tokens in the final eight language-model layers. The current screenshot, instruction, summary text, and generated action tokens remain on the frozen path. Second, a selector is trained only after the history interface is frozen. Its supervision comes from controlled restoration interventions: for an action observed in a successful trajectory, we measure how restoring a candidate event or event set changes the adapted policy’s action likelihood relative to the frozen no-image baseline.

The selector is explicitly interaction-aware. Singleton values are useful for the first choice, but are not additive: two individually useful events may be redundant, and individually weak events may be complementary. CAUSALCACHE therefore uses a lightweight set-conditioned marginal selector that repeatedly scores remaining candidates given the already selected set and may stop before exhausting the budget.

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Our contributions are:

- a policy-preserving HGKV interface that learns to use visual history while guaranteeing exact no-history parity with the frozen GUI policy;
- success-anchored restoration supervision that measures the value of restoring high-fidelity GUI events for an adapted policy;
- an HGKV-aligned two-stage selector that models conditional marginal value, event interactions, and at-most- B stopping;
- a benchmark-external training protocol in which the adapter and selector are selected only on GUI-Odyssey and evaluated zero-shot on full AndroidWorld and OS-World rosters.

Related Work

Visual GUI agents and benchmarks. Screenshot-based GUI agents increasingly learn grounding and action prediction directly from pixels, including CogAgent, SeeClick, ShowUI, and GUI-Owl (Hong et al. 2024; Cheng et al. 2024; Lin et al. 2025; Xu et al. 2026). AndroidWorld provides dynamically instantiated mobile tasks with executable success checks (Rawles et al. 2025), while OSWorld evaluates agents in real desktop applications (Xie et al. 2024). GUI-Odyssey supplies cross-application mobile trajectories and serves as our only adaptation and selector-training source (Lu et al. 2025). We use GUI-Owl-1.5-8B as the frozen cross-platform base policy (Xu et al. 2026).

Long-horizon GUI memory. GUI-Odyssey couples cross-app trajectories with a history resampler that attends to prior screenshots (Lu et al. 2025). MementoGUI learns multimodal memory writing, compression, and retrieval operators (Zeng et al. 2026). AndroTMem retrieves causally linked anchor states from interaction histories (Shi et al. 2026). These systems establish that long-horizon GUI control benefits from selective memory. CAUSALCACHE focuses on a complementary bottleneck: before selecting historical images, the policy must be trained to interpret them without losing its original action capabilities.

Interventional memory value. Recent work studies memory through controlled interventions or active state influence (Srivastava 2026; Liu et al. 2026). Our intervention is narrower and executable: every event always retains the same low-fidelity record, while a selected event additionally exposes exactly one archived post-action image. The resulting score is a policy-specific behavioral surrogate rather than an environment-level causal effect. Coalition-conditioned marginals are related to cooperative-game attribution (Shapley 1953), but our goal is not to estimate a generic Shapley value; it is to train a deployable budgeted selector for a fixed GUI policy.

Problem Formulation

At decision t , the agent receives instruction g , current observation x_t , and history

$$H_t = \{e_1, \dots, e_{t-1}\}, \quad e_j = (x_j, a_j, x_{j+1}, \delta_j), \quad (1)$$

where δ_j is an optional deterministic UI delta. Each event has a compact low-fidelity record $\tilde{e}_j$ and an archived high-fidelity post-action screenshot v_j .

For a candidate set $S \subseteq \{1, \dots, t-1\}$, let $X_t(S)$ denote a mixed-fidelity prompt that includes $\tilde{e}_j$ for every event and additionally includes v_j only for $j \in S$. The current screenshot is always included. The selector chooses

$$M_t \subseteq \{1, \dots, t-1\}, \quad |M_t| \leq B, \quad (2)$$

where B is the number of restored historical screenshots. Unselected events are not deleted; they remain available through their summaries and external archive.

We separate two goals. The history interface should improve the action policy when useful visual evidence is present while preserving the frozen policy at $B = 0$. Given that interface, the selector should maximize executable success under a fixed budget. Because full closed-loop counterfactual labels are prohibitively expensive, selector training uses a dense success-anchored action surrogate and closed-loop benchmarks provide final validation.

CausalCache

Policy-Preserving History-Gated KV Adapter

Let h_l be the input to layer l 's attention projections. In the last eight language-model layers, CAUSALCACHE adds rank-8, $\alpha = 16$ low-rank residuals (Hu et al. 2022) only to key and value projections:

$$\begin{aligned} K_l &= W_l^K h_l + M_{\text{hist}} \Delta W_l^K h_l, \\ V_l &= W_l^V h_l + M_{\text{hist}} \Delta W_l^V h_l. \end{aligned} \quad (3)$$

M_{hist} is a token mask that is true only for restored historical image tokens. It excludes instruction tokens, event-summary text, the current observation image, and action tokens. The mask is constructed fail-closed from the multimodal token-type sequence and per-image patch geometry; any mismatch between declared images and encoded image blocks aborts the forward pass.

When no history image is restored, the adapter context is absent and the projection hook returns the original module output without computing a residual. Thus

$$\pi_{\text{HG}}(\cdot \mid X_t(\emptyset)) = \pi_0(\cdot \mid X_t(\emptyset)) \quad (4)$$

up to bitwise equality in our implementation. During autoregressive generation, the adapter acts only during the prompt prefill; its historical K/V increments are then carried by the ordinary KV cache, while decode tokens bypass the adapter.

Counterfactual History Supervision

For a successful GUI-Odyssey decision with target action a_t^* , we construct four prompts with the same target:

- **correct**: the selected event summaries and images are aligned;
- **B0**: no historical screenshot is restored;
- **shuffled**: the same restored slots receive permuted images;
- **irrelevant**: restored slots receive donor images from another trajectory.

Let $\ell_c, \ell_0, \ell_s, \ell_i$ be the mean target-token log likelihoods under these variants, where ℓ_0 is computed by the frozen policy and detached. We train only the HGKV parameters with

$$\begin{aligned}\mathcal{L}_{\text{gain}} &= [m - (\ell_c - \ell_0)]_+, \\ \mathcal{L}_{\text{rank}} &= [m - (\ell_c - \ell_s)]_+ + [m - (\ell_c - \ell_i)]_+, \\ \mathcal{L}_{\text{ignore}} &= \text{SmoothL1}(\ell_s, \ell_0) + \text{SmoothL1}(\ell_i, \ell_0), \\ \mathcal{L} &= \mathcal{L}_{\text{gain}} + \mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{ignore}} + \lambda \|\Delta W\|_2^2.\end{aligned}\quad (5)$$

We use $m = 0.01$, $\lambda = 10^{-4}$, and no ordinary action cross-entropy. This prevents GUI-Odyssey from becoming a new action-imitation curriculum for the base policy; it supplies only comparative supervision about when historical visual evidence should help or be ignored.

Success-Anchored Set Utility

After freezing the history adapter, we define the utility of a restored set S as

$$U_t(S) = \frac{1}{L} \sum_{k=1}^L \log \pi_{\text{HG}}(a_{t,k}^* \mid a_{t,<k}^*, X_t(S)) - \ell_{0,t}. \quad (6)$$

Because of exact B0 parity, $\ell_{0,t}$ may be computed by either the frozen or history-gated policy under $X_t(\emptyset)$. The conditional marginal of candidate j given an already selected set S is

$$\Delta_t(j \mid S) = U_t(S \cup \{j\}) - U_t(S). \quad (7)$$

Unlike a fixed event score, Eq. 7 preserves redundancy and complementarity. Every utility used for formal selector evaluation is obtained by rerunning the complete selected set; we never report a sum of singleton gains as the utility of a multi-event coalition.

HGKV Counterfactual Readout Features

The selector should use the same interface through which the policy reads history. For each candidate and each adapted layer, we capture the frozen decision query Q_l at the final prompt token, the candidate’s frozen history keys and values ($K_{l,j}^0, V_{l,j}^0$), and the adapter increments ($\Delta K_{l,j}, \Delta V_{l,j}$). We then compute a differential readout

$$\begin{aligned}\Delta r_{l,j} &= \text{Attn}(Q_l, K_{l,j}^0 + \Delta K_{l,j})(V_{l,j}^0 + \Delta V_{l,j}) \\ &\quad - \text{Attn}(Q_l, K_{l,j}^0)V_{l,j}^0.\end{aligned}\quad (8)$$

This calculation is taken at the projection outputs where the adapter is injected. It is a lightweight counterfactual feature, not an exact replay of the full normalized and rotary-positioned attention block. For each layer we concatenate the mean-head readout difference and per-head norms, producing 160 features; eight layers yield a 1,280-dimensional candidate representation. Event features can be cached after extraction, while the current decision is encoded once per state.

Two-Stage Budgeted Selector

Stage 1: singleton scorer. A shared $160 \rightarrow 64$ layer encoder with learned attention over the eight adapted layers maps each candidate to an embedding. Four heads predict

Algorithm 1 At-most- B HGKV Set-Conditioned Selection

Require: Candidate features E , budget B

```

1:  $S \leftarrow \emptyset$ 
2: Score first events and STOP with Stage 1
3: if STOP outranks every event then
4:   return  $S$ 
5: end if
6: Add the highest-ranked event to  $S$ 
7: while  $|S| < B$  do
8:   Predict candidate ranks, marginals, and STOP with Stage 2
9:   if STOP outranks every remaining event then
10:    break
11:  end if
12:  Add the highest-ranked remaining event to  $S$ 
13: end while
14: return  $S$ 
```

singleton gain, within-state rank, positive gain, and a state-level STOP score. Training combines SmoothL1 regression, pairwise logistic ranking, positive-label binary cross entropy, and STOP binary cross entropy. Stage 1 selects the first event, supplies the independent singleton-gain baseline, and initializes Stage 2. For $B = 1$, the singleton and set-conditioned paths are identical: both compare the same Stage-1 candidate ranks with the same STOP score.

Stage 2: set-conditioned marginal scorer. Permutation-invariant set models commonly aggregate elements through pooled or attention-based encoders (Zaheer et al. 2017; Lee et al. 2019). Our Stage 2 uses a deliberately lightweight variant: candidate embeddings already encode the current decision through their HGKV readouts, one self-attention block contextualizes selected-event embeddings, and one cross-attention block maps remaining candidates to that selected context. A remaining-budget embedding conditions marginal, rank, positive, and STOP heads. It is trained from real conditional edges such as $U(\{i, j\}) - U(\{i\})$ and $U(\{i, j, k\}) - U(\{i, j\})$, generated along fixed shortlist, recent, greedy, and small-beam paths rather than by enumerating the full power set.

At inference, Stage 1 makes the first choice; Stage 2 then repeatedly compares the best remaining candidate rank with STOP. Stage 2 re-scores all remaining candidates in the capped inventory; Stage-1 shortlists are used only to control conditional-label construction cost: The deployed V1 selector supports $B \in \{1, 2, 4\}$. We separately use $B = 8$ as a large-memory policy stress test.

Experimental Setup

Training and Model Selection

All adapter and selector parameters are trained and selected only on corrected GUI-Odyssey trajectories (Lu et al. 2025). Coordinates are interpreted in their native normalized range; system-button actions and terminal states are retained, and incomplete trajectories are excluded. Splits are trajectory-disjoint. The history adapter is selected only from GUI-

Odyssey history-use metrics. The Stage-1 and Stage-2 selectors use train-episode GroupKFold for epoch selection, followed by a full-train refit and one read of the GUI-Odyssey development set. No AndroidWorld or OSWorld task is used to choose an architecture, checkpoint, loss, threshold, or budget.

Policies and Memory Baselines

The frozen policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026). We compare:

- **Frozen:** no trainable policy adapter;
- **Full-layer LoRA:** the strongest corrected GUI-Odyssey baseline, modifying all language-model Q/K/V/O projections (Hu et al. 2022);
- **Top-8 ungated KV:** an architecture-matched control using the same layers, K/V targets, rank, and alpha as HGKV but without token gating;
- **HGKV:** the proposed policy-preserving history interface.

Memory baselines are Random, Recent, OCR/RGB Similarity, the Stage-1 marginal selector, and the Stage-2 set-conditioned selector. Similarity averages OCR token-set Jacard and a joint RGB-histogram cosine score. Every method receives the same low-fidelity history and the same number of restored images.

Zero-Shot Benchmarks and Metrics

AndroidWorld (Rawles et al. 2025) is evaluated on all 116 task templates with two fixed dynamic instances per template. We average the two instances within each template and report template-macro success, with 10,000 template-clustered paired bootstrap replicates. The primary headline includes all templates; a predefined 107-template action-compatible view is a sensitivity analysis. OSWorld (Xie et al. 2024) uses the fixed 361-task no-Google-Drive roster and reports mean normalized task score. Infrastructure failures remain in an intention-to-treat denominator unless the environment fails before any policy step under a policy-agnostic void rule.

For policy adaptation we fix the selector to Recent and report $B \in \{0, 1, 2, 4, 8\}$, with the mean over positive budgets. For selection we hold HGKV fixed and report $B \in \{1, 2, 4\}$, with a separate comparison between a selected set at $B = 4$ and Recent-8. Offline selector metrics include true selected-set utility, recovery relative to an exact or declared approximate oracle, regret, STOP rate, and latency.

Results

HGKV Learns History Use without B0 Drift

Table 1 reports the frozen GUI-Odyssey gate used to select $hg-s100$. The frozen policy benefits from the presence of recent history on average but assigns higher likelihood to shuffled or irrelevant images than to the aligned history. HGKV increases the correct-history gain, reverses both content comparisons, and leaves B0 logits exactly unchanged. The small correct-shuffled margin and the positive drift of wrong histories also motivate a selector that increases the density of useful evidence.

Held-out contrast (nats/token)	Frozen	HGKV
Correct – B0	+0.0502	+0.1372
Correct – shuffled	−0.0142	+0.0016
Correct – irrelevant	−0.0115	+0.0108
B0 maximum absolute logit drift	–	0.000000

Table 1: History-use gate on 1,220 paired GUI-Odyssey development groups. All three HGKV contrast intervals have positive 95% bootstrap lower bounds; the B0 path is bitwise identical to the frozen policy.

Zero-Shot Policy Use and Memory Selection

Table 2 factorizes the end-to-end comparison into two questions. Panel A fixes memory to Recent and asks whether a policy learns to use increasing amounts of history. The comparison between Top-8 ungated KV and HGKV tests whether gains arise from adding capacity or from restricting adaptation to history tokens. Its B0 column is especially diagnostic: HGKV must equal Frozen by construction, whereas ordinary adapters may alter the current-screen policy. Panel B instead fixes HGKV and asks which history to restore. It excludes $B = 8$, which is outside the selector’s training and deployment contract, and reports a direct selected-4 versus Recent-8 contrast.

Why conditional marginals matter. The singleton-label quality gate contains 1,621 GUI-Odyssey development states and 11,364 real candidates. The exact best singleton exceeds Recent-1 by 0.0205 nats/token with 95% CI [0.0187, 0.0223], establishing learnable headroom beyond recency. However, 73.4% of states have positive singleton utility for every candidate, while only 9.3% have no positive candidate. This explains why independent positive-gain selection tends to fill the budget and why later choices require coalition-conditioned marginal labels and STOP. Every end-to-end selector cell in Table 2 uses real closed-loop execution. The corresponding offline gate reruns each complete selected set through HGKV rather than summing singleton labels.

Selector Utility, Stopping, and Cost

Table 3 reports policy-level mechanism metrics on GUI-Odyssey development states. Utility is Eq. 6 for the complete selected set. Exact search is used for small candidate sets; longer histories use a declared approximate oracle. The formal selected-set gate requires the set-conditioned selector to beat Recent, Similarity, and Random at each $B \in \{1, 2, 4\}$ with a positive paired bootstrap lower bound.

Ablations and Analyses

The ablations follow the same factorization as Table 2 rather than forming a policy-selector Cartesian product. Policy-use ablations hold Recent memory fixed, compare Frozen, full-layer LoRA, ungated top-8 KV, and HGKV, and retain $B = 8$ as a large-history stress test. Selection ablations hold HGKV fixed and compare Recent, Similarity, Random, singleton marginal, and set-conditioned selection only at $B \in \{1, 2, 4\}$. Additional analyses include adapter depth and

Panel A: Learning to use history with fixed Recent memory

Benchmark	Policy adaptation	$B = 0$	$B = 1$	$B = 2$	$B = 4$	$B = 8$	Avg. $B > 0$
AndroidWorld	Frozen GUI-Owl	TBD	TBD	TBD	TBD	TBD	TBD
	Full-layer LoRA	TBD	TBD	TBD	TBD	TBD	TBD
	Top-8 ungated KV	TBD	TBD	TBD	TBD	TBD	TBD
	HGKV (ours)	= Frozen	TBD	TBD	TBD	TBD	TBD
OSWorld	Frozen GUI-Owl	TBD	TBD	TBD	TBD	TBD	TBD
	Full-layer LoRA	TBD	TBD	TBD	TBD	TBD	TBD
	Top-8 ungated KV	TBD	TBD	TBD	TBD	TBD	TBD
	HGKV (ours)	= Frozen	TBD	TBD	TBD	TBD	TBD

Panel B: Learning which history to select with fixed HGKV

Benchmark	Memory selector	$B = 1$	$B = 2$	$B = 4$	Avg. $B=1-4$	Δ vs. Recent-8
AndroidWorld	Recent	TBD	TBD	TBD	TBD	—
	Marginal (Stage 1)	TBD	TBD	TBD	TBD	TBD
	Set-conditioned (ours)	= Marginal	TBD	TBD	TBD	TBD
OSWorld	Recent	TBD	TBD	TBD	TBD	—
	Marginal (Stage 1)	TBD	TBD	TBD	TBD	TBD
	Set-conditioned (ours)	= Marginal	TBD	TBD	TBD	TBD

Table 2: Zero-shot end-to-end results. Panel A averages $B \in \{1, 2, 4, 8\}$; Panel B averages $B \in \{1, 2, 4\}$ and defines the final column as the same-benchmark score of the $B = 4$ selected set minus Recent-8. The two learned selectors share the Stage-1 first choice, so their $B = 1$ results are identical. AndroidWorld reports template-macro success (%); OSWorld reports mean normalized task score (%). All adapters and selectors are trained and selected only on GUI-Odyssey.

Selector	$U(B=1)$	$U(B=2)$	$U(B=4)$	Oracle recovery	Regret	STOP rate	Latency
Random	TBD	TBD	TBD	TBD	TBD	TBD	—
Recent	TBD	TBD	TBD	TBD	TBD	TBD	—
Similarity	TBD	TBD	TBD	TBD	TBD	TBD	TBD
HGKV singleton marginal	TBD	TBD	TBD	TBD	TBD	TBD	TBD
HGKV set-conditioned	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Oracle / approximate oracle	TBD	TBD	TBD	1.000	0	—	—

Table 3: Selector mechanism analysis. Multi-event utility is always measured by rerunning the complete coalition; it is never approximated by adding singleton gains.

rank, removal of gain/rank/ignore losses, selected top- B versus bottom- B , wrong-event replacement, summary-image mismatch, and temporal reversal. We also stratify results by trajectory length, app, and the gap between Recent and oracle memory.

Limitations and Claim Boundary

The success-anchored utility in Eq. 6 is a dense teacher-forced surrogate, not a direct long-horizon return. Its value must ultimately be judged by closed-loop success. The word “causal” in CAUSALCACHE refers only to a controlled intervention that restores selected screenshots while holding the remaining prompt fixed; we do not identify environment-level structural causality or learn a world model.

Our zero-shot claim concerns CAUSALCACHE adaptation: neither the adapter nor the selector uses AndroidWorld or OSWorld data for training or model selection. We cannot guarantee that the pretrained base policy never encountered related data. GUI-Odyssey is mobile-only, so OSWorld evaluates whether the learned history interface transfers beyond

its adaptation platform. The current HGKV adapter still raises the likelihood of some wrong-history variants, and the set-conditioned selector is partly motivated by this limited content margin. Finally, the current selector supports budgets through four restored events; $B = 8$ is evaluated only as a policy-level memory stress test.

Conclusion

CAUSALCACHE separates learning to use visual history from learning which history to show. A token-gated KV adapter adds a history interface to a frozen GUI policy without changing its no-history behavior. Success-anchored restoration labels then supervise a selector whose later choices condition on the already selected set and may stop early. The resulting design isolates the value of historical evidence from generic policy fine-tuning and enables one GUI-Odyssey-trained system to be evaluated zero-shot across mobile and desktop environments. **[Pending: Update the conclusion with final AndroidWorld and OSWorld findings.]**

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